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Bitwise Operators

A Bit of Fun

Sommario

- Bitwise operators
- Reason why
- Single & multiple bits operations
- Shift operators

Bitwise Operators

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- Memory is composed by “bits”
- Every data is then stored in memory in binary
- Usually we do no need to access the single bit
- C anyway provides us specific operators that work at bit level

Reason why

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- Why we should deal with single bits?
  - Access to raw HW often requires to deal with single bits
  - More efficient arithmetic operations
  - More efficient storing of data

Bitwise operators

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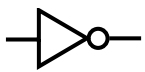
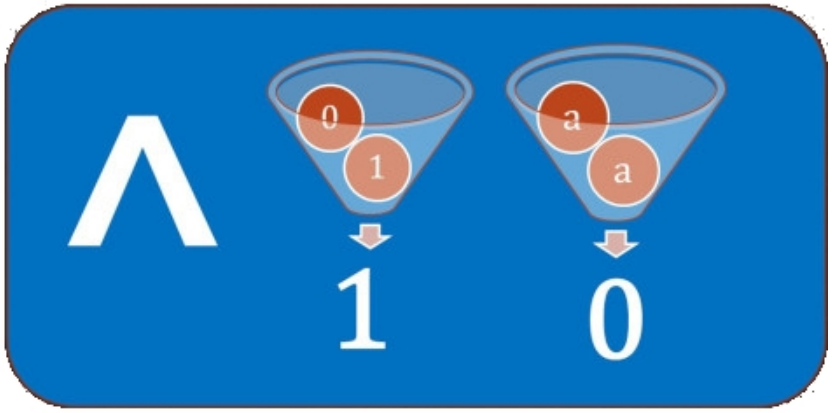
- They can work on single bits
  - `&` → binary AND 
  - `|` → binary OR 
  - `~` → binary NOT 
  - `^` → binary XOR 
  - `<<` → left shift
  - `>>` → right shift

Single bit operations



Multiple bits operations

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- These operators are applied to integer types only!
  - Usually multiple bits!
- The operator is applied to the corresponding bits in each data

AND

```
  0110
& 1100
------
  0100
```

OR

```
  0110
| 1100
------
  1110
```

XOR

```
  0110
^ 1100
------
  1010
```

NOT

```
~ 1100
------
  0011
```

- Note that results are different from the same logical operator!

Shift operators

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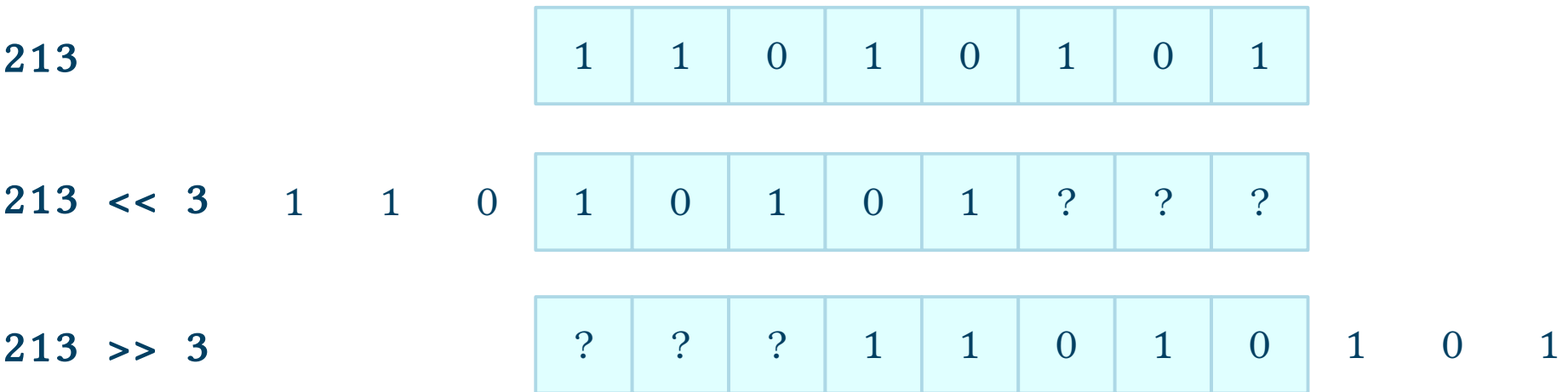
- Shift all the bits of a data to the right or to the left
- Look at the following example
- We want to shift 213 by 3



Shift operators

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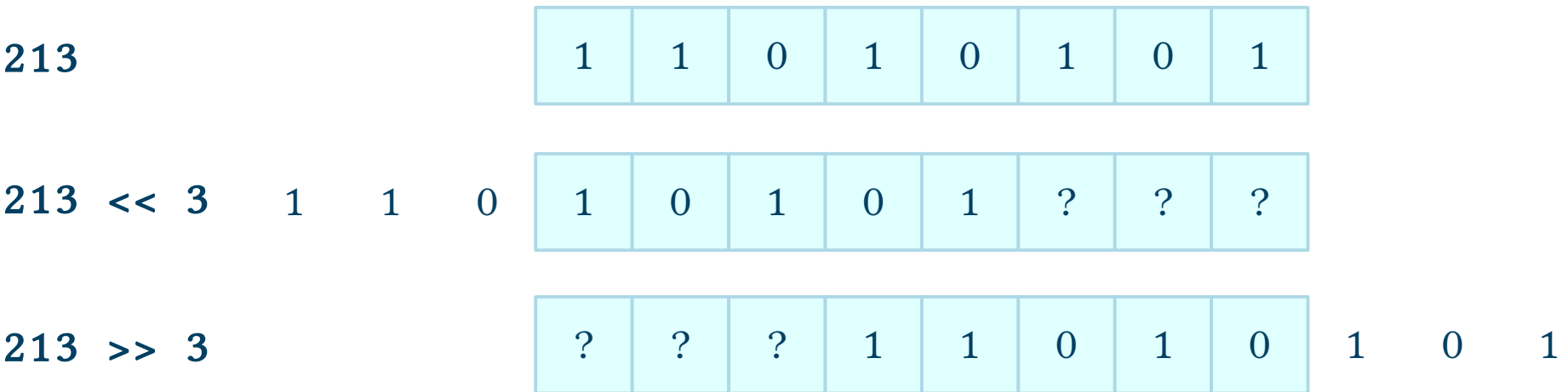
- In both cases
  - Bits shifted off the end are lost



Shift operators

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- Not symmetrical behavior
  - Left shift → new lower order bits are filled in with 0s
  - Right shift → highest order bit is replicated

`213`	1	1	0	1	0	1	0	1
`213 << 3`	1	0	1	0	1	0	0	0
`213 >> 3`	1	1	1	1	1	0	1	0

Biwise Operator Tricks

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- | with 1 is useful for turning select bits on
- & with 0 is useful for turning select bits off
- | is useful for taking the union of bits
- & is useful for taking the intersection of bits
- ^ is useful for flipping specific bits
- ~ is useful for flipping all bits

Practical usage

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- Integer number manipulation
  - Get/extract a specific byte of an integer or specific bits
  - e.g., complement letter case
- Bit masking for flags/configuration etc.
- Fast math
  - Left/right shifts are multiplication/divide by two
  - Trick for power of 2 check

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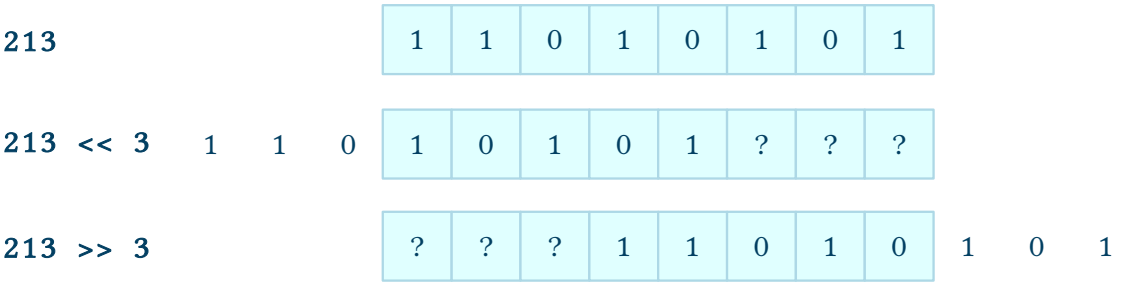

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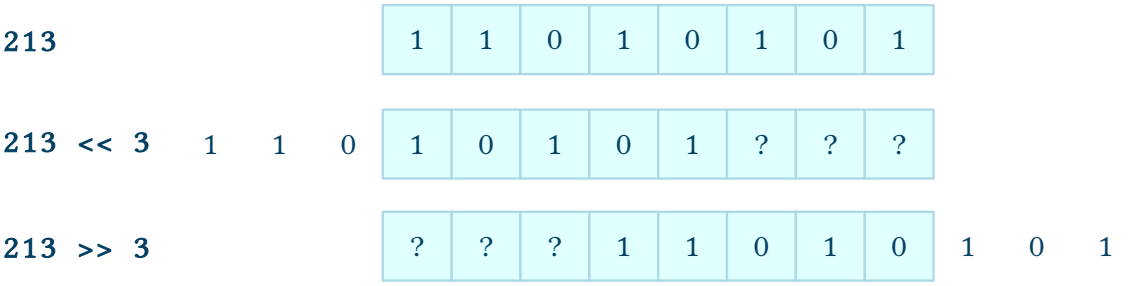

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`213`

1	1	0	1	0	1	0	1

`213 << 3`

1	0	1	0	1	0	0	0

`213 >> 3`

1	1	1	1	1	0	1	0

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